Quantum Computing

Lecture 2: More Quantum Theory
Deutsch's Algorithm

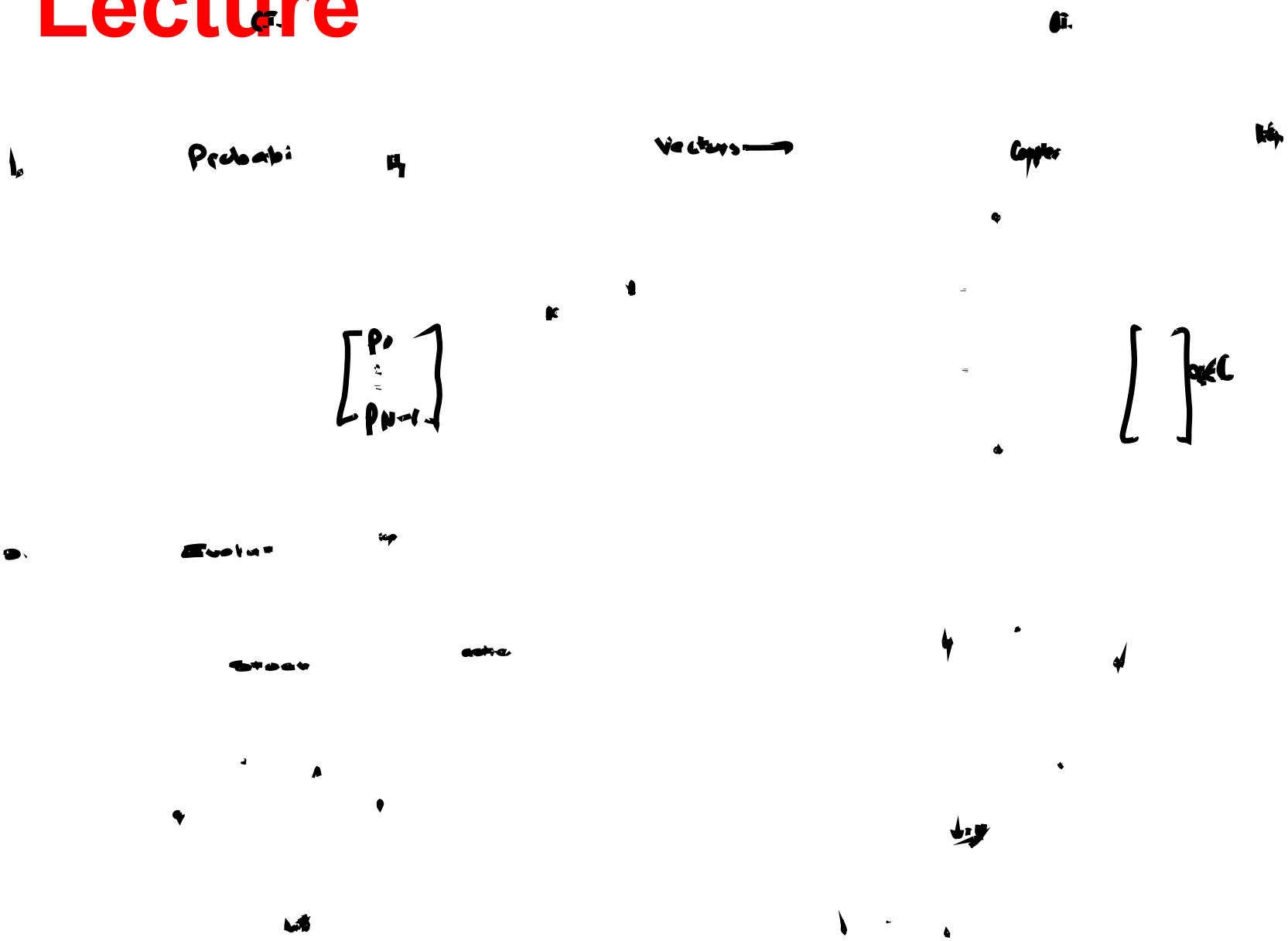
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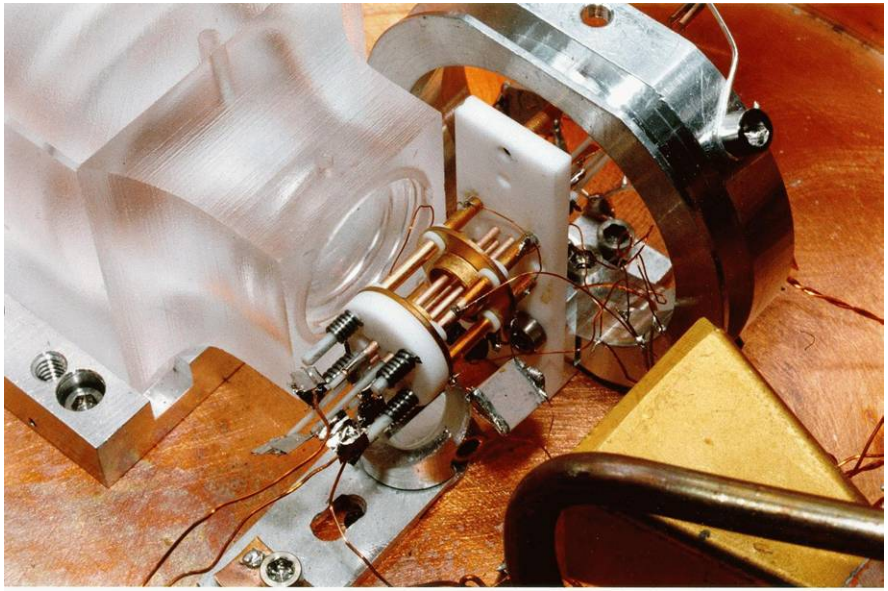
Summary of Last Lecture



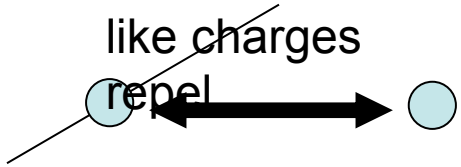
Summary of Last Lecture

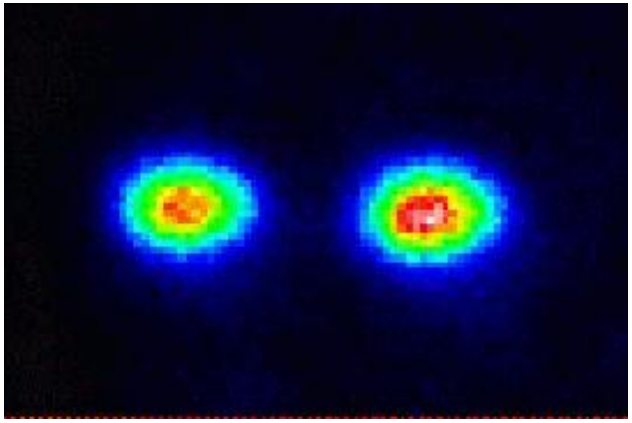


Ion Trap

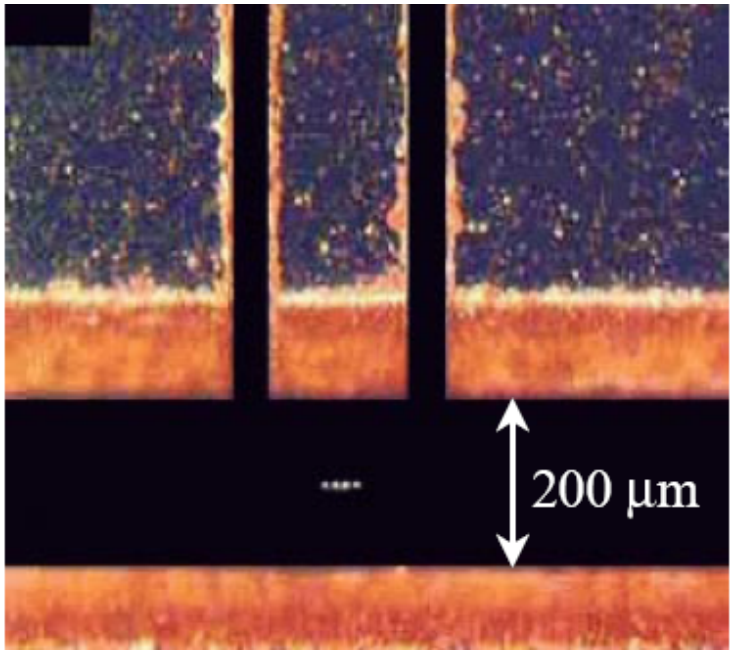


Oscillating electric fields trap ions

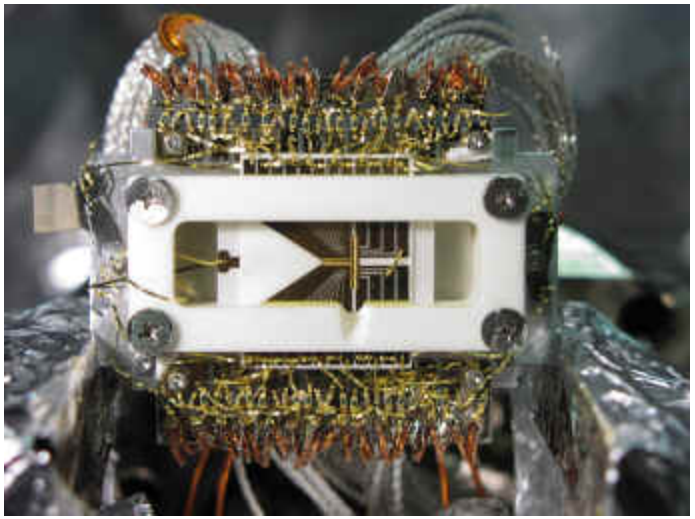


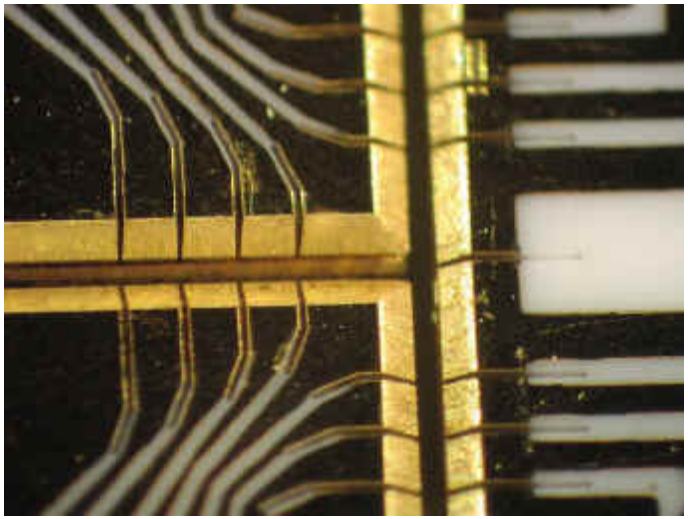


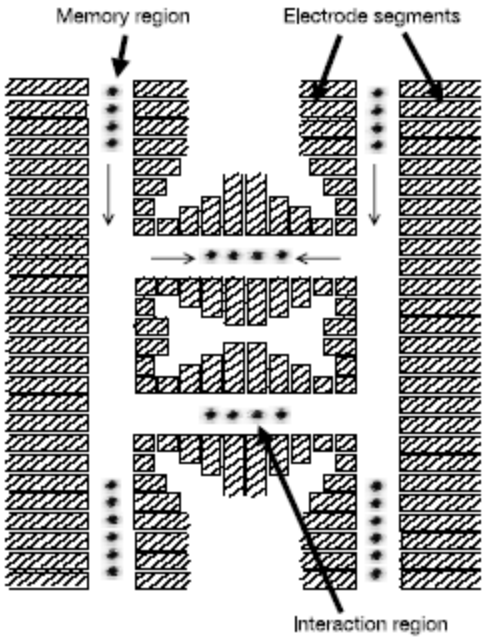
2 9Be+ Ions in an Ion Trap

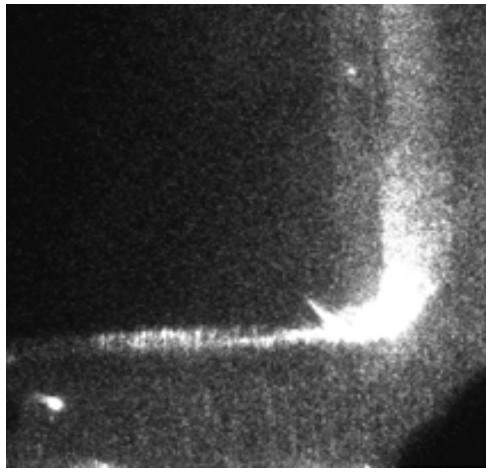


Shuttling Around a Corner









Pictures snatched from Chris Monroe's University of Michigan website

Qubits

Two dimensional quantum systems are called qubits

A qubit has a wave function which we write as

$$|v\rangle = v_0|0\rangle + v_1|1\rangle \qquad |v_0|^2 + |v_1|^2 = 1$$

Examples:

Valid qubit wave functions:

$$|v\rangle = |0\rangle \qquad |||v\rangle|| = \sqrt{|1|^2 + |0|^2} = 1$$

$$|v\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle \quad |||v\rangle|| = \sqrt{\left|\frac{1}{\sqrt{2}}\right|^2 + \left|\frac{i}{\sqrt{2}}\right|^2} = 1$$

Invalid qubit wave function (not normalized):

$$|v\rangle = 5|0\rangle + i|1\rangle \qquad |||v\rangle|| = \sqrt{|5|^2 + |i|^2} = \sqrt{26}$$

Measuring Qubits

A bit is a classical system with two possible states, 0 and 1

A qubit is a quantum system with two possible states, 0 and 1

When we observe a qubit, we get the result 0 or the result 1

0	or	1

If before we observe the qubit the wave function of the qubit is

$$|v\rangle = v_0|0\rangle + v_1|1\rangle$$

then the probability that we observe 0 is $|v_0|^2$

and the probability that we observe 1 is $|v_1|^2$

"measuring in the computational basis"

Measuring Qubits

Example:

We are given a qubit with wave function

$$|v\rangle = \frac{1}{\sqrt{3}}|0\rangle + i\sqrt{\frac{2}{3}}|1\rangle \quad |||v\rangle|| = \sqrt{\left|\frac{1}{\sqrt{3}}\right|^2 + \left|i\sqrt{\frac{2}{3}}\right|^2} = 1$$

If we observe the system in the computational basis, then we get outcome 0 with probability

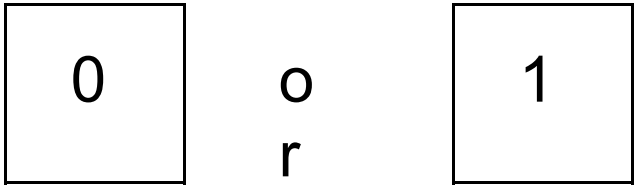
$$\left|\frac{1}{\sqrt{3}}\right|^2 = \frac{1}{3}$$

and we get outcome 1 with probability:

$$\left|\frac{\sqrt{2}i}{\sqrt{3}}\right|^2 = \frac{2}{3}$$

Measuring Qubits Continued

When we observe a qubit, we get the result 0 or the result 1



If before we observe the qubit the wave function of the qubit is

$$|v\rangle = v_0|0\rangle + v_1|1\rangle$$

then the probability that we observe 0 is $|v_0|^2$

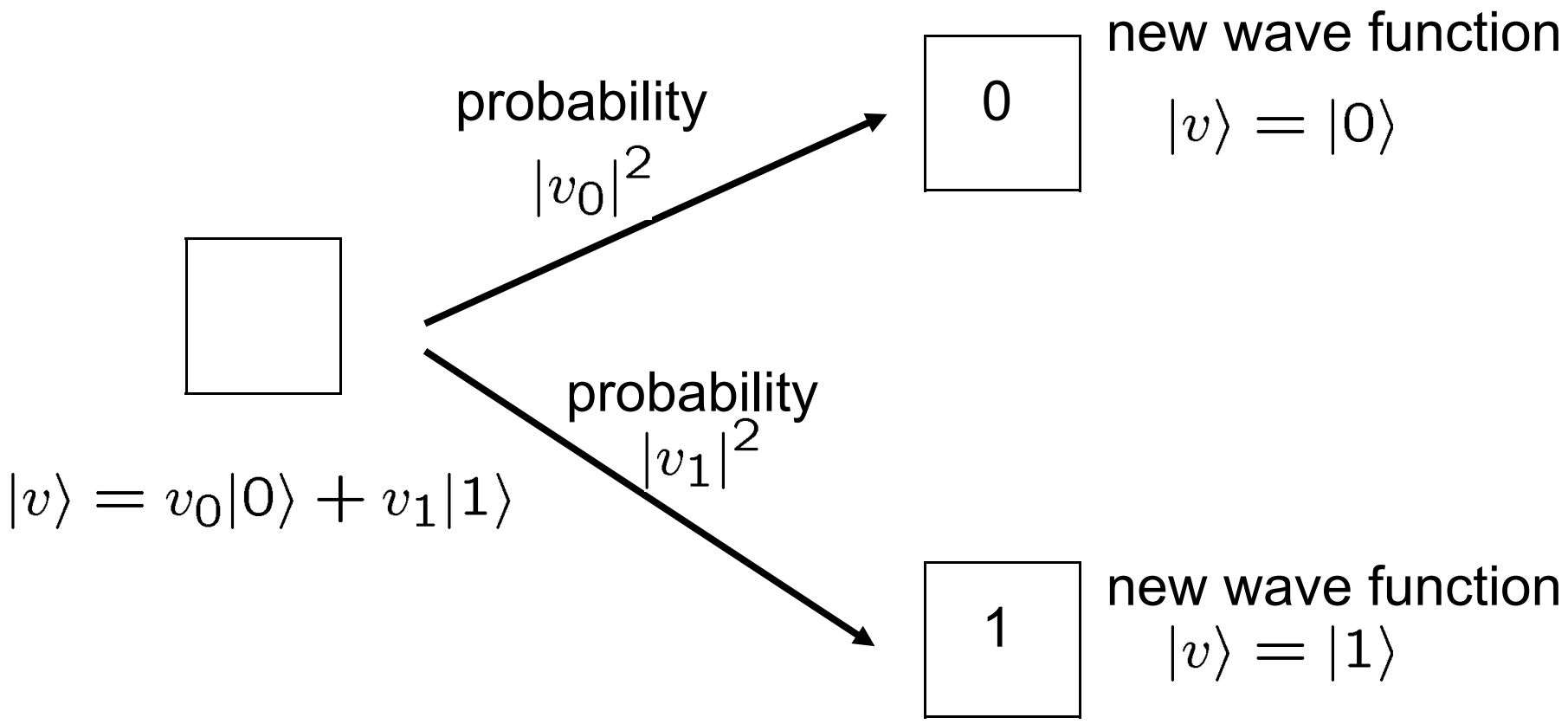
and the new wave function for the qubit is $|0\rangle$

and the probability that we observe 1 is $|v_1|^2$

and the new wave function for the qubit is $|1\rangle$

"measuring in the computational basis"

Measuring Qubits Continued



The wave function is a description of our system.

When we measure the system we find the system in one state

This happens with probabilities we get from our description

Measuring Qubits

Example:

We are given a qubit with wave function

$$|v\rangle = \frac{1}{\sqrt{3}}|0\rangle + i\sqrt{\frac{2}{3}}|1\rangle \quad |||v\rangle|| = \sqrt{\left|\frac{1}{\sqrt{3}}\right|^2 + \left|i\sqrt{\frac{2}{3}}\right|^2} = 1$$

If we observe the system in the computational basis, then we get outcome 0 with probability

$$\left|\frac{1}{\sqrt{3}}\right|^2 = \frac{1}{3}$$ new wave function $|0\rangle$

and we get outcome 1 with probability:

$$\left|\frac{2i}{\sqrt{3}}\right|^2 = \frac{2}{3}$$ new wave function $|1\rangle$

Measuring Qubits

Example:

We are given a qubit with wave function

$$|v\rangle = |0\rangle \qquad |||v\rangle|| = \sqrt{|1|^2 + |0|^2} = 1$$

If we observe the system in the computational basis, then we get outcome 0 with probability

$$|1|^2 = 1$$ new wave function $|0\rangle$

and we get outcome 1 with probability:

$$|0|^2 = 0$$ a.k.a never

Unitary Evolution for Qubits

Unitary evolution will be described by a two dimensional unitary matrix

$$U = \begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix}$$

If initial qubit wave function is

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \end{bmatrix} = v_0|0\rangle + v_1|1\rangle$$

Then this evolves to

$$|v'\rangle = \begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix} \begin{bmatrix} v_0 \\ v_1 \end{bmatrix} = \begin{bmatrix} U_{0,0}v_0 + U_{0,1}v_1 \\ U_{1,0}v_0 + U_{1,1}v_1 \end{bmatrix}$$

Unitary Evolution for Qubits

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix} \qquad |v\rangle = \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix} = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$$

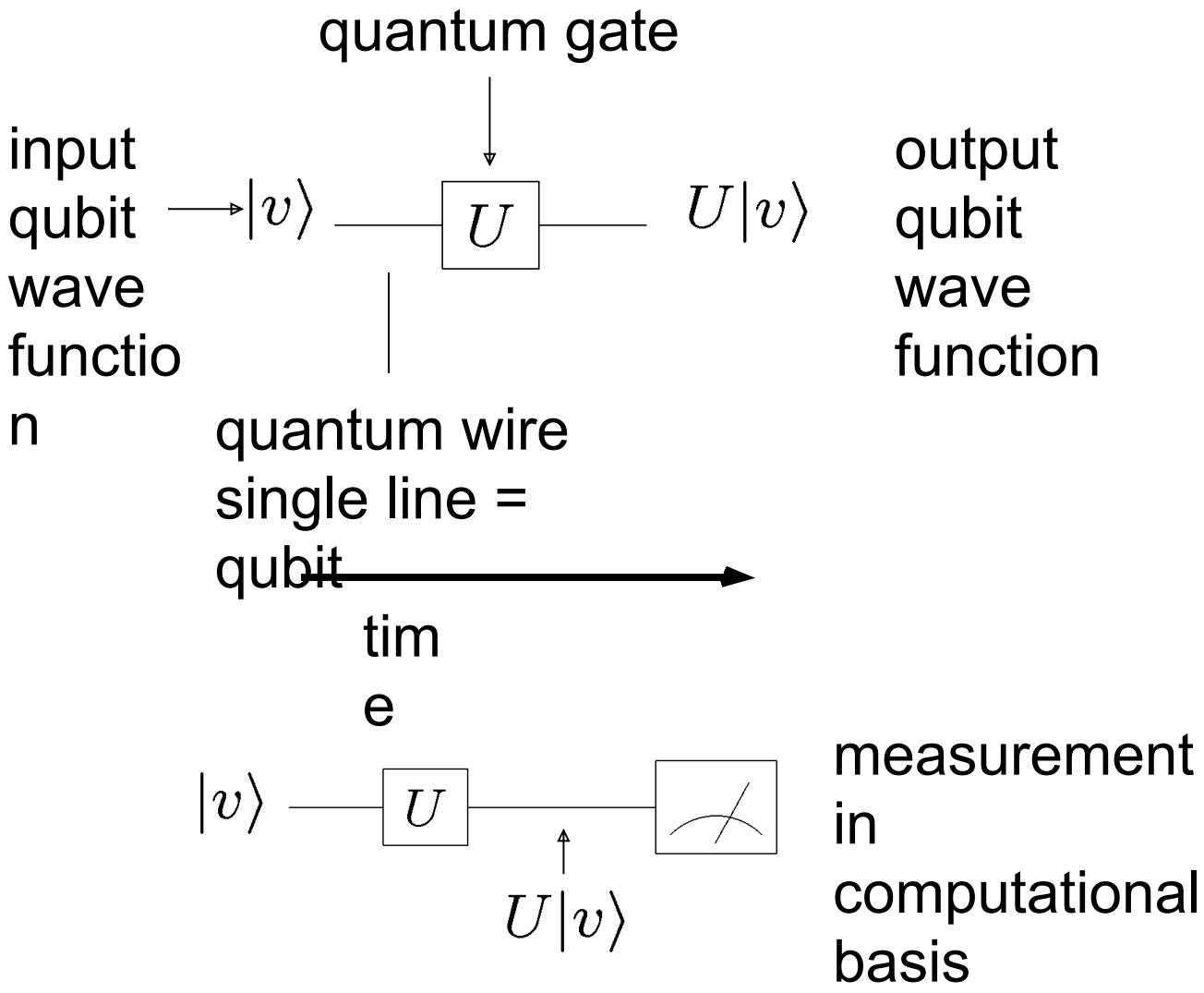
$$|v'\rangle = U|v\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}}\frac{1}{2} + \frac{1}{\sqrt{2}}\frac{\sqrt{3}}{2} \\ \frac{-i}{\sqrt{2}}\frac{1}{2} + \frac{i}{\sqrt{2}}\frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} \frac{1+\sqrt{3}}{2\sqrt{2}} \\ \frac{(-1+\sqrt{3})i}{2\sqrt{2}} \end{bmatrix}$$

$$|v'\rangle = \frac{1+\sqrt{3}}{2\sqrt{2}}|0\rangle + \frac{(-1+\sqrt{3})i}{2\sqrt{2}}|1\rangle$$

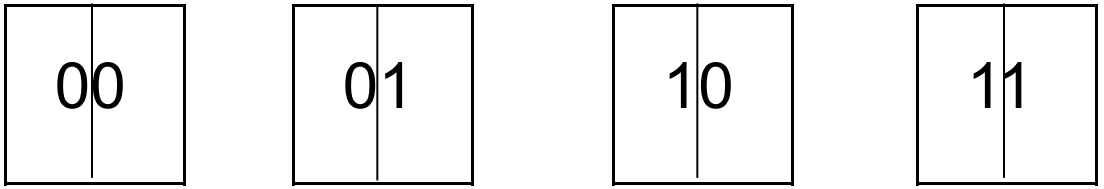
Single Qubit Quantum Circuits

Circuit diagrams for evolving qubits

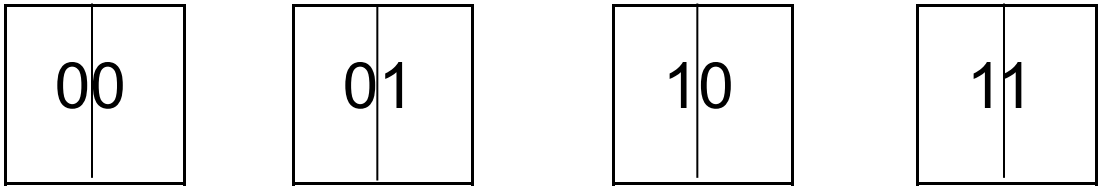


Two Qubits

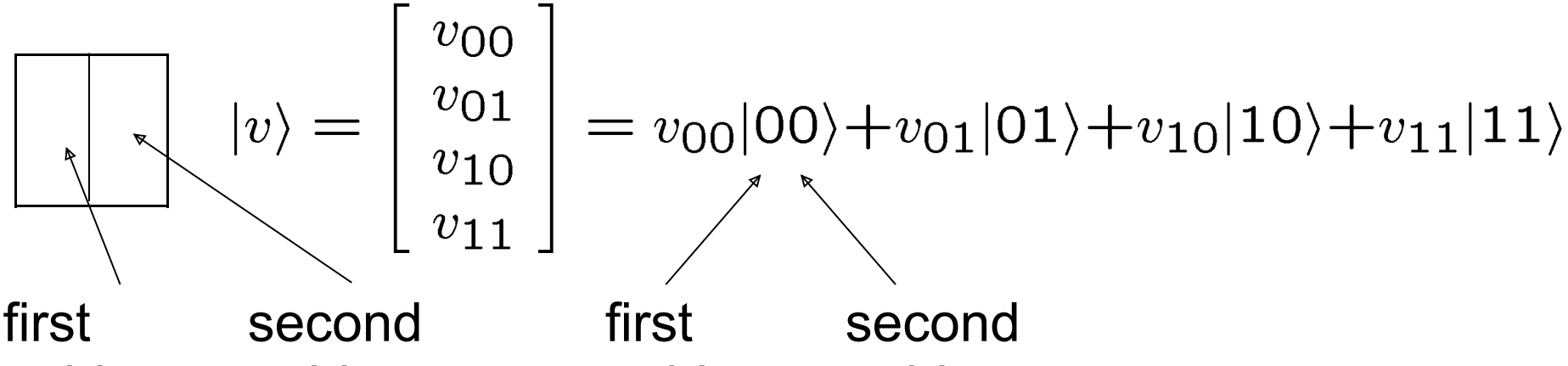
Two bits can be in one of four different states



Similarly two qubits have four different states



The wave function for two qubits thus has four components:



Two Qubits

Examples:

$$|v\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \\ 0 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}|00\rangle + \frac{i}{\sqrt{2}}|01\rangle$$

$$|v\rangle = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{\sqrt{3}}{2} \end{bmatrix} = \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|11\rangle$$

$$|v\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = |00\rangle$$

When Two Qubits Are Two

The wave function for two qubits has four components:

$$|v\rangle = \begin{bmatrix} v_{00} \\ v_{01} \\ v_{10} \\ v_{11} \end{bmatrix} = v_{00}|00\rangle + v_{01}|01\rangle + v_{10}|10\rangle + v_{11}|11\rangle$$

Sometimes we can write the wave function of two qubits as the "tensor product" of two one qubit wave functions.

$|v\rangle = |a\rangle \otimes |b\rangle$ "separable"

$$|a\rangle = a_0|0\rangle + a_1|1\rangle \qquad |b\rangle = b_0|0\rangle + b_1|1\rangle$$

$$|v\rangle = (a_0|0\rangle + a_1|1\rangle) \otimes (b_0|0\rangle + b_1|1\rangle)$$

$$= a_0b_0|0\rangle\otimes|0\rangle + a_0b_1|0\rangle\otimes|1\rangle + a_1b_0|1\rangle\otimes|0\rangle + a_1b_1|1\rangle\otimes|1\rangle$$

$$= a_0b_0|00\rangle + a_0b_1|01\rangle + a_1b_0|10\rangle + a_1b_1|11\rangle$$

Two Qubits, Separable

Example : $|v\rangle = |a\rangle \otimes |b\rangle$

$$|a\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle \quad |b\rangle = \frac{i}{\sqrt{5}}|0\rangle + \frac{2}{\sqrt{5}}|1\rangle$$

$$|v\rangle = |a\rangle \otimes |b\rangle = \left(\frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle\right) \otimes \left(\frac{i}{\sqrt{5}}|0\rangle + \frac{2}{\sqrt{5}}|1\rangle\right)$$

$$= \frac{1}{2}\frac{i}{\sqrt{5}}|0\rangle \otimes |0\rangle + \frac{1}{2}\frac{2}{\sqrt{5}}|0\rangle \otimes |1\rangle + \frac{\sqrt{3}}{2}\frac{i}{\sqrt{5}}|1\rangle \otimes |0\rangle + \frac{\sqrt{3}}{2}\frac{2}{\sqrt{5}}|1\rangle \otimes |1\rangle$$

$$= \frac{i}{2\sqrt{5}}|00\rangle + \frac{1}{\sqrt{5}}|01\rangle + \frac{\sqrt{3}i}{2\sqrt{5}}|10\rangle + \frac{\sqrt{3}}{\sqrt{5}}|11\rangle$$

Two Qubits, Entangled

Example:

$$|v\rangle = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{\sqrt{3}}{2} \end{bmatrix} = \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|11\rangle$$

Assume

$|v\rangle = |a\rangle \otimes |b\rangle$ $\quad |a\rangle = a_0|0\rangle + a_1|1\rangle \quad |b\rangle = b_0|0\rangle + b_1|1\rangle$

$$= a_0b_0|00\rangle + a_0b_1|01\rangle + a_1b_0|10\rangle + a_1b_1|11\rangle$$

$a_0b_1 = 0$

Either

$a_0 = 0$ but this implies $a_0b_0 = 0$

or

$b_1 = 0$ but this implies $a_1b_1 = 0$

contradictions

So $|v\rangle$ is not a separable state. It is entangled

Measuring Two Qubits

If we measure both qubits in the computational basis, then we get one of four outcomes: 00, 01, 10, and 11

If the wave function for the two qubits is

$$|v\rangle = \begin{bmatrix} v_{00} \\ v_{01} \\ v_{10} \\ v_{11} \end{bmatrix} = v_{00}|00\rangle + v_{01}|01\rangle + v_{10}|10\rangle + v_{11}|11\rangle$$

Probability of 00 is $|v_{00}|^2$ New wave function is $|00\rangle$

Probability of 01 is $|v_{01}|^2$ New wave function is $|01\rangle$

Probability of 10 is $|v_{10}|^2$ New wave function is $|10\rangle$

Probability of 11 is $|v_{11}|^2$ New wave function is $|11\rangle$

Two Qubits, Measuring

Example:

$$|v\rangle = \frac{i}{2\sqrt{5}}|00\rangle + \frac{1}{\sqrt{5}}|01\rangle + \frac{\sqrt{3}i}{2\sqrt{5}}|10\rangle + \frac{\sqrt{3}}{\sqrt{5}}|11\rangle$$

Probability of 00 is $|v_{00}|^2 = \left|\frac{i}{2\sqrt{5}}\right|^2 = \frac{1}{20}$

Probability of 01 is $|v_{01}|^2 = \left|\frac{1}{\sqrt{5}}\right|^2 = \frac{1}{5}$

Probability of 10 is $|v_{10}|^2 = \left|\frac{\sqrt{3}i}{2\sqrt{6}}\right|^2 = \frac{3}{20}$

Probability of 11 is $|v_{11}|^2 = \left|\frac{\sqrt{3}}{\sqrt{5}}\right|^2 = \frac{3}{5}$

Two Qubit Evolutions

Rule 2: The wave function of a N dimensional quantum system evolves in time according to a unitary matrix U. If the wave function initially is $|v\rangle$ then after the evolution correspond to U the new wave function is $|v'\rangle = U|v\rangle$

$$U = \begin{bmatrix} U_{00,00} & U_{00,01} & U_{00,10} & U_{00,11} \\ U_{01,00} & U_{01,01} & U_{01,10} & U_{01,11} \\ U_{10,00} & U_{10,01} & U_{10,10} & U_{10,11} \\ U_{11,00} & U_{11,01} & U_{11,10} & U_{11,11} \end{bmatrix} \qquad |v\rangle = \begin{bmatrix} v_{00} \\ v_{01} \\ v_{10} \\ v_{11} \end{bmatrix}$$

$$|v'\rangle = U|v\rangle$$

$$|v'\rangle = \begin{bmatrix} U_{00,00}v_{00} + U_{00,01}v_{01} + U_{00,10}v_{10} + U_{00,11}v_{11} \\ U_{01,00}v_{00} + U_{01,01}v_{01} + U_{01,10}v_{10} + U_{01,11}v_{11} \\ U_{10,00}v_{00} + U_{10,01}v_{01} + U_{10,10}v_{10} + U_{10,11}v_{11} \\ U_{11,00}v_{00} + U_{11,01}v_{01} + U_{11,10}v_{10} + U_{11,11}v_{11} \end{bmatrix}$$

Two Qubit Evolutions

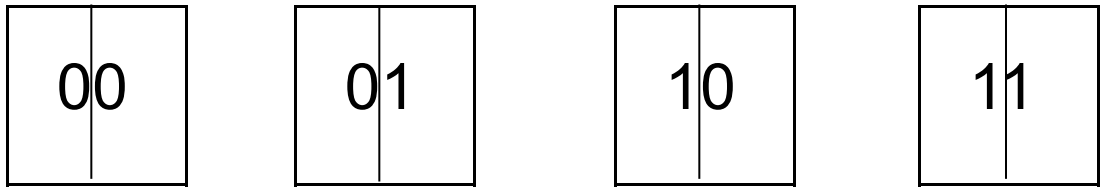
$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 & 0 \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad |v\rangle = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{\sqrt{3}}{2} \end{bmatrix} = \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|11\rangle$$

$$|v'\rangle = \begin{bmatrix} U_{00,00}v_{00} + U_{00,01}v_{01} + U_{00,10}v_{10} + U_{00,11}v_{11} \\ U_{01,00}v_{00} + U_{01,01}v_{01} + U_{01,10}v_{10} + U_{01,11}v_{11} \\ U_{10,00}v_{00} + U_{10,01}v_{01} + U_{10,10}v_{10} + U_{10,11}v_{11} \\ U_{11,00}v_{00} + U_{11,01}v_{01} + U_{11,10}v_{10} + U_{11,11}v_{11} \end{bmatrix}$$

$$|v'\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}}\frac{1}{2} + \frac{i}{\sqrt{2}} \cdot 0 + 0 \cdot 0 + 0 \cdot \frac{\sqrt{3}}{2} \\ \frac{i}{\sqrt{2}}\frac{1}{2} + \frac{1}{\sqrt{2}} \cdot 0 + 0 \cdot 0 + 0 \cdot \frac{\sqrt{3}}{2} \\ 0 \cdot \frac{1}{2} + 0 \cdot 0 + 0 \cdot 0 + 1 \cdot \frac{\sqrt{3}}{2} \\ 0 \cdot \frac{1}{2} + 0 \cdot 0 + 1 \cdot 0 + 0 \cdot \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2\sqrt{2}} \\ \frac{i}{2\sqrt{2}} \\ \frac{\sqrt{3}}{2} \\ 0 \end{bmatrix}$$

Manipulations of Two Bits

Two bits can be in one of four different states



We can manipulate these bits

$$00 \longrightarrow 01$$
$$01 \longrightarrow 00$$
$$10 \longrightarrow 10$$
$$11 \longrightarrow 11$$

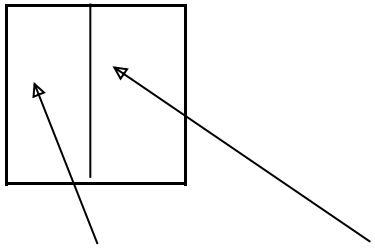
Sometimes this can be thought of as just operating on one of the bits (for example, flip the second bit):

$$00 \longrightarrow 01$$
$$01 \longrightarrow 00$$
$$10 \longrightarrow 11$$
$$11 \longrightarrow 10$$

But sometimes we cannot (as in the first example above)

Manipulations of Two Qubits

Similarly, we can apply unitary operations on only one of the qubits at a time:



first qubit

second qubit

Unitary operator that acts only on the first qubit

$$U = V \otimes I$$

two dimensional unitary matrix

two dimensional Identity matrix

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Unitary operator that acts only on the second qubit:

$$U = I \otimes V$$

Tensor Product of Matrices

$$U = V \otimes W \qquad V = \begin{bmatrix} V_{0,0} & V_{0,1} \\ V_{1,0} & V_{1,1} \end{bmatrix} \quad W = \begin{bmatrix} W_{0,0} & W_{0,1} \\ W_{1,0} & W_{1,1} \end{bmatrix}$$

$$U = \begin{bmatrix} V_{0,0} & V_{0,1} \\ V_{1,0} & V_{1,1} \end{bmatrix} \otimes \begin{bmatrix} W_{0,0} & W_{0,1} \\ W_{1,0} & W_{1,1} \end{bmatrix}$$

$$U = \begin{bmatrix} V_{0,0}W_{0,0} & V_{0,0}W_{0,1} & V_{0,1}W_{0,0} & V_{0,1}W_{0,1} \\ V_{0,0}W_{1,0} & V_{0,0}W_{1,1} & V_{0,1}W_{1,0} & V_{0,1}W_{1,1} \\ V_{1,0}W_{0,0} & V_{1,0}W_{0,1} & V_{1,1}W_{0,0} & V_{1,1}W_{0,1} \\ V_{1,0}W_{1,0} & V_{1,0}W_{1,1} & V_{1,1}W_{1,0} & V_{1,1}W_{1,1} \end{bmatrix}$$

Tensor Product of Matrices

Example: $U = V \otimes W$

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \quad W = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$$

$$U = \begin{bmatrix} V_{0,0}W_{0,0} & V_{0,0}W_{0,1} & V_{0,1}W_{0,0} & V_{0,1}W_{0,1} \\ V_{0,0}W_{1,0} & V_{0,0}W_{1,1} & V_{0,1}W_{1,0} & V_{0,1}W_{1,1} \\ V_{1,0}W_{0,0} & V_{1,0}W_{0,1} & V_{1,1}W_{0,0} & V_{1,1}W_{0,1} \\ V_{1,0}W_{1,0} & V_{1,0}W_{1,1} & V_{1,1}W_{1,0} & V_{1,1}W_{1,1} \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}}W_{0,0} & \frac{1}{\sqrt{2}}W_{0,1} & \frac{i}{\sqrt{2}}W_{0,0} & \frac{i}{\sqrt{2}}W_{0,1} \\ \frac{1}{\sqrt{2}}W_{1,0} & \frac{1}{\sqrt{2}}W_{1,1} & \frac{i}{\sqrt{2}}W_{1,0} & \frac{i}{\sqrt{2}}W_{1,1} \\ \frac{i}{\sqrt{2}}W_{0,0} & \frac{i}{\sqrt{2}}W_{0,1} & \frac{1}{\sqrt{2}}W_{0,0} & \frac{1}{\sqrt{2}}W_{0,1} \\ \frac{i}{\sqrt{2}}W_{1,0} & \frac{i}{\sqrt{2}}W_{1,1} & \frac{1}{\sqrt{2}}W_{1,0} & \frac{1}{\sqrt{2}}W_{1,1} \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}}\frac{1}{2} & \frac{1}{\sqrt{2}}\frac{\sqrt{3}}{2} & \frac{i}{\sqrt{2}}\frac{1}{2} & \frac{i}{\sqrt{2}}\frac{\sqrt{3}}{2} \\ \frac{1}{\sqrt{2}}\frac{\sqrt{3}}{2} & \frac{1}{\sqrt{2}}\frac{-1}{2} & \frac{i}{\sqrt{2}}\frac{\sqrt{3}}{2} & \frac{i}{\sqrt{2}}\frac{-1}{2} \\ \frac{i}{\sqrt{2}}\frac{1}{2} & \frac{i}{\sqrt{2}}\frac{\sqrt{3}}{2} & \frac{1}{\sqrt{2}}\frac{1}{2} & \frac{1}{\sqrt{2}}\frac{\sqrt{3}}{2} \\ \frac{i}{\sqrt{2}}\frac{\sqrt{3}}{2} & \frac{i}{\sqrt{2}}\frac{-1}{2} & \frac{1}{\sqrt{2}}\frac{\sqrt{3}}{2} & \frac{1}{\sqrt{2}}\frac{-1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2\sqrt{2}} & \frac{\sqrt{3}}{2\sqrt{2}} & \frac{i}{2\sqrt{2}} & \frac{\sqrt{3}i}{2\sqrt{2}} \\ \frac{\sqrt{3}}{2\sqrt{2}} & \frac{-1}{2\sqrt{2}} & \frac{\sqrt{3}i}{2\sqrt{2}} & \frac{-i}{2\sqrt{2}} \\ \frac{i}{2\sqrt{2}} & \frac{\sqrt{3}i}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & \frac{\sqrt{3}}{2\sqrt{2}} \\ \frac{\sqrt{3}i}{2\sqrt{2}} & \frac{-i}{2\sqrt{2}} & \frac{\sqrt{3}}{2\sqrt{2}} & \frac{-1}{2\sqrt{2}} \end{bmatrix}$$

Tensor Product of Matrices

Example: $U = V \otimes I$

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \qquad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} V_{0,0}W_{0,0} & V_{0,0}W_{0,1} & V_{0,1}W_{0,0} & V_{0,1}W_{0,1} \\ V_{0,0}W_{1,0} & V_{0,0}W_{1,1} & V_{0,1}W_{1,0} & V_{0,1}W_{1,1} \\ V_{1,0}W_{0,0} & V_{1,0}W_{0,1} & V_{1,1}W_{0,0} & V_{1,1}W_{0,1} \\ V_{1,0}W_{1,0} & V_{1,0}W_{1,1} & V_{1,1}W_{1,0} & V_{1,1}W_{1,1} \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}}1 & \frac{1}{\sqrt{2}}0 & \frac{i}{\sqrt{2}}1 & \frac{i}{\sqrt{2}}0 \\ \frac{1}{\sqrt{2}}0 & \frac{1}{\sqrt{2}}1 & \frac{i}{\sqrt{2}}0 & \frac{i}{\sqrt{2}}1 \\ \frac{i}{\sqrt{2}}1 & \frac{i}{\sqrt{2}}0 & \frac{1}{\sqrt{2}}1 & \frac{1}{\sqrt{2}}0 \\ \frac{i}{\sqrt{2}}0 & \frac{i}{\sqrt{2}}1 & \frac{1}{\sqrt{2}}0 & \frac{1}{\sqrt{2}}1 \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{i}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{i}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Tensor Product of Matrices

Example: $U = I \otimes W$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad W = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$U = \begin{bmatrix} V_{0,0}W_{0,0} & V_{0,0}W_{0,1} & V_{0,1}W_{0,0} & V_{0,1}W_{0,1} \\ V_{0,0}W_{1,0} & V_{0,0}W_{1,1} & V_{0,1}W_{1,0} & V_{0,1}W_{1,1} \\ V_{1,0}W_{0,0} & V_{1,0}W_{0,1} & V_{1,1}W_{0,0} & V_{1,1}W_{0,1} \\ V_{1,0}W_{1,0} & V_{1,0}W_{1,1} & V_{1,1}W_{1,0} & V_{1,1}W_{1,1} \end{bmatrix}$$

$$U = \begin{bmatrix} 1\frac{1}{\sqrt{2}} & 1\frac{i}{\sqrt{2}} & 0\frac{1}{\sqrt{2}} & 0\frac{i}{\sqrt{2}} \\ 1\frac{i}{\sqrt{2}} & 1\frac{1}{\sqrt{2}} & 0\frac{i}{\sqrt{2}} & 0\frac{1}{\sqrt{2}} \\ 0\frac{1}{\sqrt{2}} & 0\frac{i}{\sqrt{2}} & 1\frac{1}{\sqrt{2}} & 1\frac{i}{\sqrt{2}} \\ 0\frac{i}{\sqrt{2}} & 0\frac{1}{\sqrt{2}} & 1\frac{i}{\sqrt{2}} & 1\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 & 0 \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ 0 & 0 & \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Tensor Product of Matrices

Example: $U = I \otimes W$

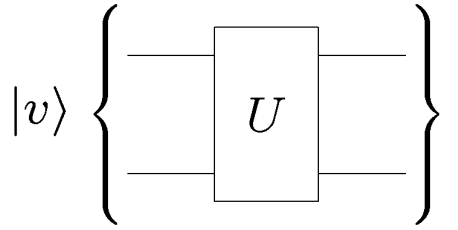
$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad W = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$U = \begin{bmatrix} V_{0,0}W_{0,0} & V_{0,0}W_{0,1} & V_{0,1}W_{0,0} & V_{0,1}W_{0,1} \\ V_{0,0}W_{1,0} & V_{0,0}W_{1,1} & V_{0,1}W_{1,0} & V_{0,1}W_{1,1} \\ V_{1,0}W_{0,0} & V_{1,0}W_{0,1} & V_{1,1}W_{0,0} & V_{1,1}W_{0,1} \\ V_{1,0}W_{1,0} & V_{1,0}W_{1,1} & V_{1,1}W_{1,0} & V_{1,1}W_{1,1} \end{bmatrix}$$

$$U = \begin{bmatrix} 1\frac{1}{\sqrt{2}} & 1\frac{i}{\sqrt{2}} & 0\frac{1}{\sqrt{2}} & 0\frac{i}{\sqrt{2}} \\ 1\frac{i}{\sqrt{2}} & 1\frac{1}{\sqrt{2}} & 0\frac{i}{\sqrt{2}} & 0\frac{1}{\sqrt{2}} \\ 0\frac{1}{\sqrt{2}} & 0\frac{i}{\sqrt{2}} & 1\frac{1}{\sqrt{2}} & 1\frac{i}{\sqrt{2}} \\ 0\frac{i}{\sqrt{2}} & 0\frac{1}{\sqrt{2}} & 1\frac{i}{\sqrt{2}} & 1\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 & 0 \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ 0 & 0 & \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

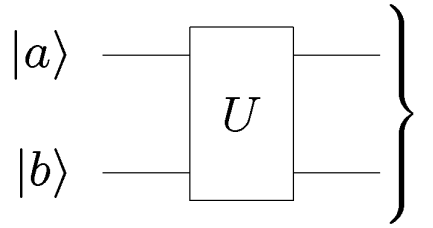
Two Qubit Quantum Circuits



$$|v'\rangle = U|v\rangle$$

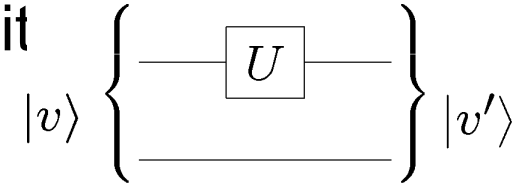
A two qubit unitary gate

Sometimes the input our output is known to be seperable:

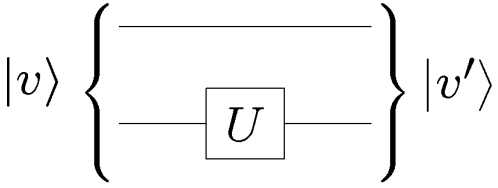


$$|v'\rangle = U(|a\rangle \otimes |b\rangle)$$

Sometimes we act only one qubit



$$|v'\rangle = (U \otimes I)|v\rangle$$

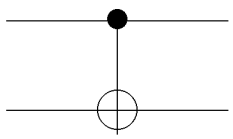


$$|v'\rangle = (I \otimes U)|v\rangle$$

Some Two Qubit Gates

control

target



controlled-NOT $= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

$$|00\rangle \rightarrow |00\rangle$$

$$|01\rangle \rightarrow |01\rangle$$

$$|10\rangle \rightarrow |11\rangle$$

$$|11\rangle \rightarrow |10\rangle$$

Conditional on the first bit, the gate flips the second bit.

Computational Basis and Unitaries

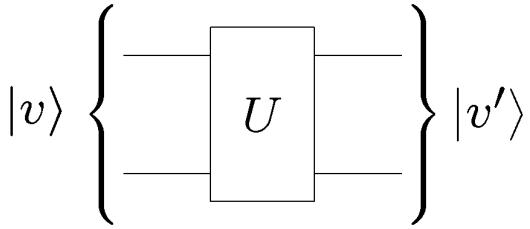
$$U = \begin{bmatrix} U_{00,00} & U_{00,01} & U_{00,10} & U_{00,11} \\ U_{01,00} & U_{01,01} & U_{01,10} & U_{01,11} \\ U_{10,00} & U_{10,01} & U_{10,10} & U_{10,11} \\ U_{11,00} & U_{11,01} & U_{11,10} & U_{11,11} \end{bmatrix} \qquad |00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$|v'\rangle = U|00\rangle = v'_{00}|00\rangle + v'_{01}|01\rangle + v'_{10}|10\rangle + v'_{11}|11\rangle$$

$$|v'\rangle = \begin{bmatrix} U_{00,00}v_{00} + U_{00,01}v_{01} + U_{00,10}v_{10} + U_{00,11}v_{11} \\ U_{01,00}v_{00} + U_{01,01}v_{01} + U_{01,10}v_{10} + U_{01,11}v_{11} \\ U_{10,00}v_{00} + U_{10,01}v_{01} + U_{10,10}v_{10} + U_{10,11}v_{11} \\ U_{11,00}v_{00} + U_{11,01}v_{01} + U_{11,10}v_{10} + U_{11,11}v_{11} \end{bmatrix}$$

Notice that by examining the unitary evolution of all computational basis states, we can explicitly determine what the unitary matrix.

Linearity



$$|v\rangle = v_{00}|00\rangle + v_{01}|01\rangle + v_{10}|10\rangle + v_{11}|11\rangle$$

$$|v'\rangle = U|v\rangle = v_{00}U|00\rangle + v_{01}U|01\rangle + v_{10}U|10\rangle + v_{11}U|11\rangle$$

We can act on each computational basis state and then resum

This simplifies calculations considerably

Linearity

Example:

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 & 0 \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \qquad |v\rangle = \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|11\rangle$$

$$|v'\rangle = U|v\rangle = \frac{1}{2}U|00\rangle + \frac{\sqrt{3}}{2}U|11\rangle$$

$$= \frac{1}{2}\left(\frac{1}{\sqrt{2}}|00\rangle + \frac{i}{\sqrt{2}}|01\rangle\right) + \frac{\sqrt{3}}{2}|10\rangle$$

$$= \frac{1}{2\sqrt{2}}|00\rangle + \frac{i}{2\sqrt{2}}|01\rangle + \frac{\sqrt{3}}{2}|10\rangle$$

Linearity

Example:

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 & 0 \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \qquad |v\rangle = \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|01\rangle$$

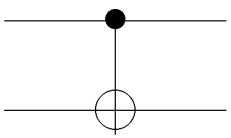
$$|v'\rangle = U|v\rangle = \frac{1}{2}U|00\rangle + \frac{\sqrt{3}}{2}U|01\rangle$$

$$= \frac{1}{2}\left(\frac{1}{\sqrt{2}}|00\rangle + \frac{i}{\sqrt{2}}|01\rangle\right) + \frac{\sqrt{3}}{2}\left(\frac{i}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|01\rangle\right)$$

$$= \frac{1 + i\sqrt{3}}{2\sqrt{2}}|00\rangle + \frac{\sqrt{3} + i}{2\sqrt{2}}|01\rangle$$

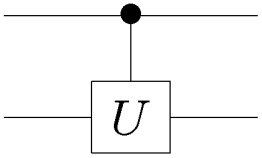
Some Two Qubit Gates

control
target

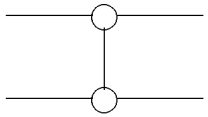


controlled-NOT $= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

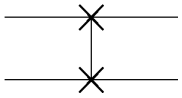
control
target



controlled-U $= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & U_{00} & U_{01} \\ 0 & 0 & U_{10} & U_{11} \end{bmatrix}$



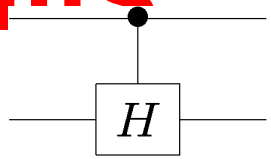
controlled-phase $= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$



swap $= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

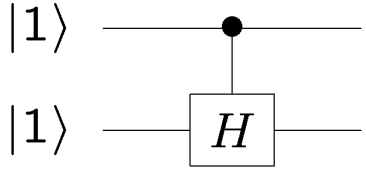
Quantum Circuits

controlled-H



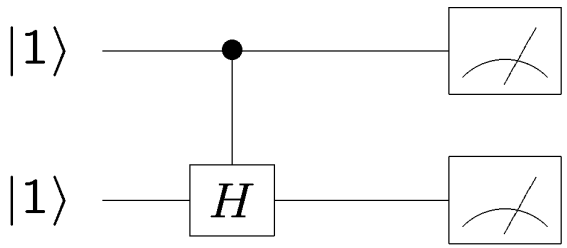
$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$|v\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = |11\rangle$$



$$|v'\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{2}}(|10\rangle - |11\rangle)$$



Probability of 10: $\left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$

Probability of 11: $\left|\frac{-1}{\sqrt{2}}\right|^2 = \frac{1}{2}$

Probability of 00 and 01: $|0|^2 = 0$

Matrices, Bras, and Kets

So far we have used bras and kets to describe row and column vectors. We can also use them to describe matrices:

Outer product of two vectors:

$$|v\rangle\langle w| = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \begin{bmatrix} w_1^* & w_2^* \end{bmatrix}$$

$$= \begin{bmatrix} v_1 w_1^* & v_1 w_2^* \\ v_2 w_1^* & v_2 w_2^* \end{bmatrix}$$

Example:

$$|v\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \qquad \langle w| = \begin{bmatrix} \frac{i}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$|v\rangle\langle w| = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{i}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}}\frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}}\frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}}\frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}}\frac{-1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{i}{2} & -\frac{1}{2} \\ \frac{i}{2} & -\frac{1}{2} \end{bmatrix}$$

Matrices, Bras, and Kets

We can expand a matrix about all of the computational basis outer products

$$M = \sum_{i,j=0}^{N-1} M_{i,j}|i\rangle\langle j| = \begin{bmatrix} M_{0,0} & \cdots & M_{N-1,0} \\ \vdots & \ddots & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

Example:

$$M = \begin{bmatrix} 1 & i \\ -1 & -i \end{bmatrix}$$

$$|0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad |0\rangle\langle 1| = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$|1\rangle\langle 0| = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \quad |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$M = |0\rangle\langle 0| + i|0\rangle\langle 1| - 1|1\rangle\langle 0| - i|1\rangle\langle 1|$$

Matrices, Bras, and Kets

We can expand a matrix about all of the computational basis outer products

$$M = \sum_{i,j=0}^{N-1} M_{i,j}|i\rangle\langle j| = \begin{bmatrix} M_{0,0} & \cdots & M_{N-1,0} \\ \vdots & \ddots & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

This makes it easy to operate on kets and bras:

$$M|v\rangle = \sum_{i,j=0}^{N-1} M_{i,j}|i\rangle\langle j|v\rangle$$

complex numbers

$$\langle w|M = \sum_{i,j=0}^{N-1} M_{i,j}\langle w|i\rangle\langle j|$$

Matrices, Bras, and Kets

Example:

$$M = \begin{bmatrix} 1 & i \\ -1 & -i \end{bmatrix}$$

$$M = |0\rangle\langle 0| + i|0\rangle\langle 1| - 1|1\rangle\langle 0| - i|1\rangle\langle 1|$$

$$|v\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$$

$$M|v\rangle = (|0\rangle\langle 0| + i|0\rangle\langle 1| - 1|1\rangle\langle 0| - i|1\rangle\langle 1|)\left(\frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle\right)$$

$$= \frac{1}{2}|0\rangle + i\frac{\sqrt{3}}{2}|0\rangle - \frac{1}{2}|1\rangle - i\frac{\sqrt{3}}{2}|1\rangle$$

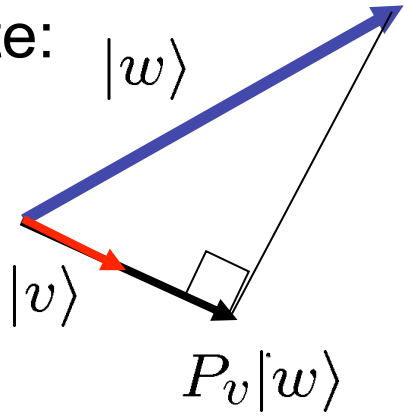
$$= \frac{1 + i\sqrt{3}}{2}|0\rangle - \frac{1 + i\sqrt{3}}{2}|1\rangle$$

Projectors

The projector onto a state $|v\rangle$ (which is of unit norm) is given by

$$P_v = |v\rangle\langle v|$$

Projects onto the state:



Note that $P_v|v\rangle = |v\rangle\langle v|v\rangle = |v\rangle$

and that

$$P_v|w\rangle = |v\rangle\langle v|w\rangle = (\langle v|w\rangle)|v\rangle$$

Example: $|v\rangle = |0\rangle \quad P_v = |0\rangle\langle 0|$

$$|w\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$$

$$P_v|w\rangle = |0\rangle\langle 0|\left(\frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle\right) = \frac{1}{2}|0\rangle$$

Measurement Rule

If we measure a quantum system whose wave function is $|v\rangle$ in the basis $|w_i\rangle$, then the probability of getting the outcome $|w_i\rangle$ corresponding to is given by

$$Pr(|w_i\rangle) = |\langle w_i|v\rangle|^2 = \langle v|w_i\rangle\langle w_i|v\rangle = \langle v|P_{w_i}|v\rangle$$

where

$$P_{w_i} = |w_i\rangle\langle w_i|$$

The new wave function of the system after getting the measurement outcome corresponding to $|w_i\rangle$ is given by

$$|v'\rangle = \frac{P_{w_i}|v\rangle}{\sqrt{Pr(|w_i\rangle)}}$$

For measuring in a complete basis, this reduces to our normal prescription for quantum measurement, but…

Measuring One of Two Qubits

Suppose we measure the first of two qubits in the computational basis. Then we can form the two projectors:

$$P_0 \otimes I = |0\rangle\langle 0| \otimes I \qquad I = |0\rangle\langle 0| + |1\rangle\langle 1|$$

$$P_1 \otimes I = |1\rangle\langle 1| \otimes I$$

If the two qubit wave function is $|v\rangle$ then the probabilities of these two outcomes are

$$Pr(0) = \langle v|P_0 \otimes I|v\rangle$$

$$Pr(1) = \langle v|P_1 \otimes I|v\rangle$$

And the new state of the system is given by either

$$|v'\rangle = \frac{P_0 \otimes I|v\rangle}{\sqrt{Pr(0)}}$$

Outcome was 0

$$|v'\rangle = \frac{P_1 \otimes I|v\rangle}{\sqrt{Pr(1)}}$$

Outcome was 1

Measuring One of Two Qubits

Example $|v\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{\sqrt{2}}|11\rangle$

Measure the first qubit: $P_0 \otimes I = |0\rangle\langle 0| \otimes I \quad P_1 \otimes I = |1\rangle\langle 1| \otimes I$

$P_0 \otimes I = |0\rangle\langle 0| \otimes (|0\rangle\langle 0| + |1\rangle\langle 1|) = |00\rangle\langle 00| + |01\rangle\langle 01|$

$$Pr(0) = \langle v|P_0 \otimes I|v\rangle$$

$$P_0 \otimes I|v\rangle = (|00\rangle\langle 00| + |01\rangle\langle 01|)|v\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle$$

$$Pr(0) = \left(\frac{1}{2}\langle 00| + \frac{1}{2}\langle 01| + \frac{1}{\sqrt{2}}\langle 11|\right)\left(\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle\right)$$

$$Pr(0) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \qquad |v'\rangle = \frac{P_0 \otimes I|v\rangle}{\sqrt{Pr(0)}} = \frac{\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle}{\frac{1}{\sqrt{2}}}$$

$$|v'\rangle = \frac{1}{\sqrt{2}}|00\rangle + \sqrt{\frac{1}{2}}|01\rangle = |0\rangle \otimes \left(\frac{1}{\sqrt{2}}|0\rangle + \sqrt{\frac{1}{2}}|1\rangle\right)$$

Instantaneous Communication?

Suppose two distant parties each have a qubit and their
joint quantum wave function is

$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

If one party now measures its qubit,
then…

$$P_0 \otimes I = |0\rangle\langle 0| \otimes I \qquad Pr(0) = \frac{1}{2} \qquad |v'\rangle = |0\rangle \otimes |0\rangle$$

$$P_1 \otimes I = |1\rangle\langle 1| \otimes I \qquad Pr(1) = \frac{1}{2} \qquad |v'\rangle = |1\rangle \otimes |1\rangle$$

The other parties qubit is now either the $|0\rangle$ or $|1\rangle$

Instantaneous communication? NO.
Why NO? These two results happen with probabilities.

Correlation does not imply communication.

Important Single Qubit Unitaries

Pauli Matrices:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

"bit flip"

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

"phase flip"

$$\sigma_0 = I \qquad \sigma_1 = X \qquad \sigma_2 = Y \qquad \sigma_3 = Z \qquad \sigma_j^2 = I$$

$|0\rangle$ — X — $|1\rangle$

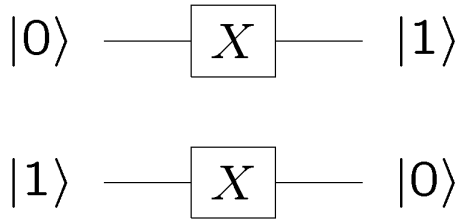
$|1\rangle$ — X — $|0\rangle$

"bit flip" is just the classical not gate

$$X\,Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} = -iZ$$

$$Y\,X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} = iZ$$

Important Single Qubit Unitaries



"bit flip" is just the classical not gate

Hadamard gate:

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \qquad H^2 = I$$



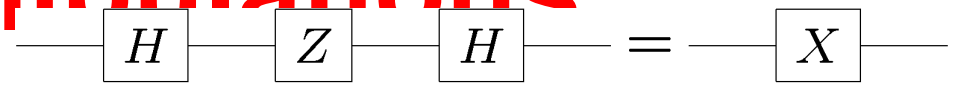
Jacques Hadamard

$$HZH = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

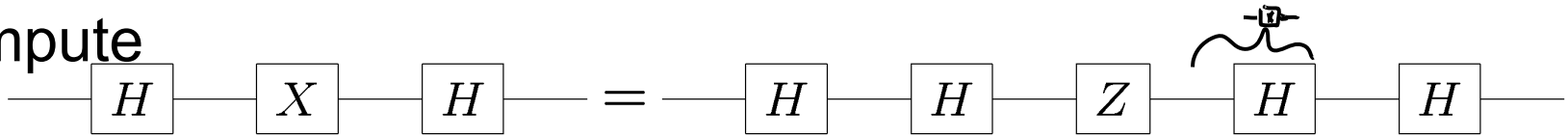
$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = X$$

$$HZH = X$$

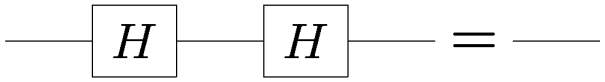
Single Qubit Manipulations



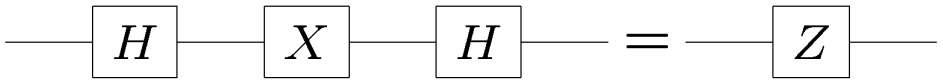
Use this to compute HXH



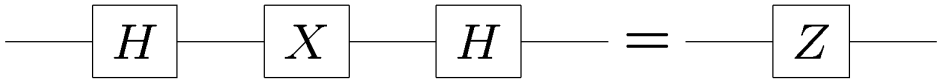
But $H^2 = I$

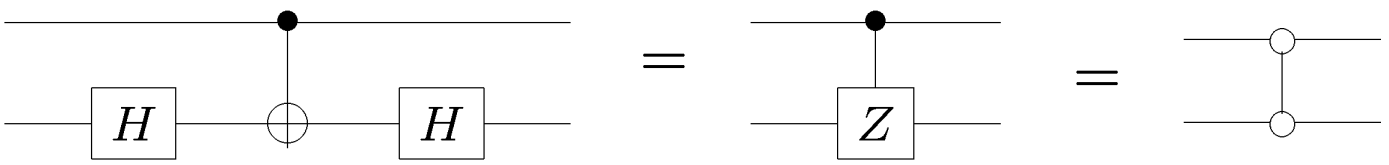


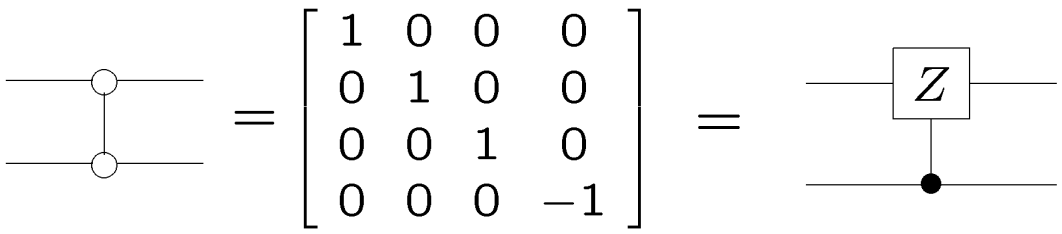
So that

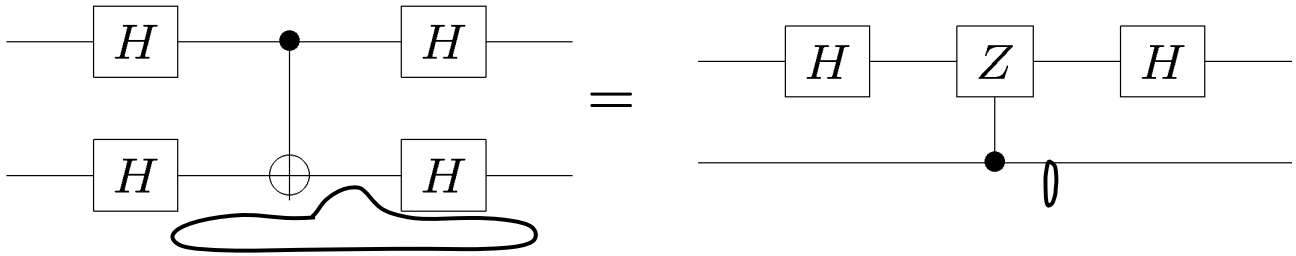


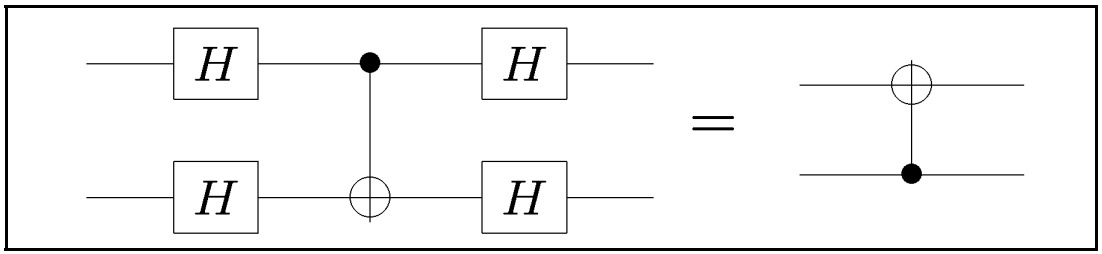
A Cool Circuit Identity

Using 



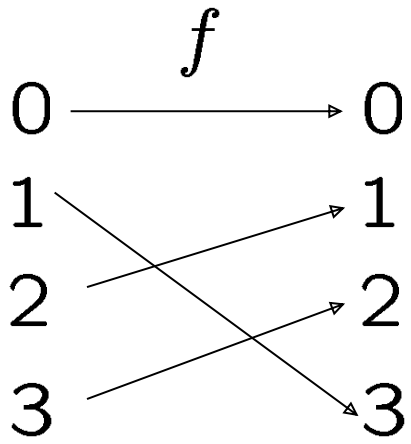






Reversible Classical Gates

A reversible classical gate on k bits is one to one function on the 2^k values of these bits.



Example :

$00 \rightarrow 00$
$01 \rightarrow 01$
$10 \rightarrow 11$
$11 \rightarrow 10$

reversibl

$00 \rightarrow 00$
$01 \rightarrow 00$
$10 \rightarrow 10$
$11 \rightarrow 11$

not

Reversible Classical Gates

A reversible classical gate on k bits is one to one function on the 2^k values of these bits.

We can represent reversible classical gates by a permutation matrix.

Permutation matrix is matrix in which every row and column contains at most one 1 and the rest of the elements are 0

Example: $00 \rightarrow 00$, $01 \rightarrow 01$, $10 \rightarrow 11$, $11 \rightarrow 10$

reversibl

$$\begin{array}{c} \\ 00 \\ 01 \\ 10 \\ 11 \end{array} \begin{array}{c} \begin{array}{cccc} 00 & 01 & 10 & 11 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

output

inpu t

Quantum Versions of Reversible Classical Gates

A reversible classical gate on k bits is one to one function on the 2^k values of these bits.

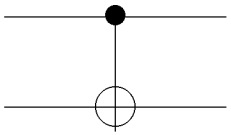
We can turn reversible classical gates into unitary quantum gates

Permutation matrix is matrix in which every row and column contains at most one 1 and the rest of the elements are 0

Use permutation matrix as unitary evolution matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

controlled-NOT



$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

David Speaks



David Deutsch 1985

"Complexity theory has been mainly concerned with constraints upon the computation of functions: which functions can be computed, how fast, and with use of how much memory. With quantum computers, as with classical stochastic computers, one must also ask 'and with what probability?' We have seen that the minimum computation time for certain tasks can be lower for Q than for T . Complexity theory for Q deserves further investigation."

Q = quantum computers
T = classical computers

Deutsch's Problem

Suppose you are given a black box which computes one of the following four reversible gates:

$00 \to 00$	$00 \to 01$	$00 \to 00$	$00 \to 01$
$01 \to 01$	$01 \to 00$	$01 \to 01$	$01 \to 00$
$10 \to 10$	$10 \to 11$	$10 \to 11$	$10 \to 10$
$11 \to 11$	$11 \to 10$	$11 \to 10$	$11 \to 11$
"identity"	NOT 2nd bit	controlled-NOT	controlled-NOT + NOT 2nd bit
	constant		balanced

Deutsch's (Classical) Problem:
How many times do we have to use this black box to determine whether we are given the first two or the second two?

Classical Deutsch's Problem

$00 \to 00$	$00 \to 01$	$00 \to 00$	$00 \to 01$
$01 \to 01$	$01 \to 00$	$01 \to 01$	$01 \to 00$
$10 \to 10$	$10 \to 11$	$10 \to 11$	$10 \to 10$
$11 \to 11$	$11 \to 10$	$11 \to 10$	$11 \to 11$
"identity"	NOT 2nd bit	controlled-NOT	controlled-NOT + NOT 2nd bit
	constant		balanced

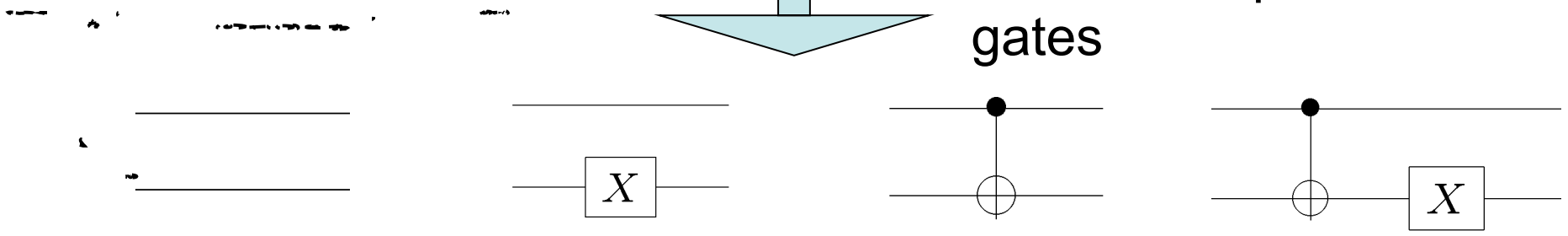
Notice that for every possible input, this does not separate the "constant" and "balanced" sets. This implies at least one use of the black box is needed.

Querying the black box with 00 and 10 distinguishes between
these two sets. Two uses of the black box are necessary and sufficient.

Classical to Quantum Deutsch

$00 \rightarrow 00$	$00 \rightarrow 01$	$00 \rightarrow 00$	$00 \rightarrow 01$
$01 \rightarrow 01$	$01 \rightarrow 00$	$01 \rightarrow 01$	$01 \rightarrow 00$
$10 \rightarrow 10$	$10 \rightarrow 11$	$10 \rightarrow 11$	$10 \rightarrow 10$
$11 \rightarrow 11$	$11 \rightarrow 10$	$11 \rightarrow 10$	$11 \rightarrow 11$
"identity"	NOT 2nd bit	controlled-NOT	controlled-NOT + NOT 2nd bit

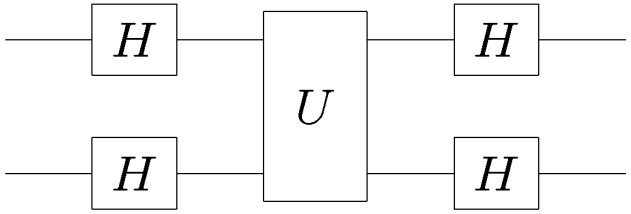
Convert to quantum gates

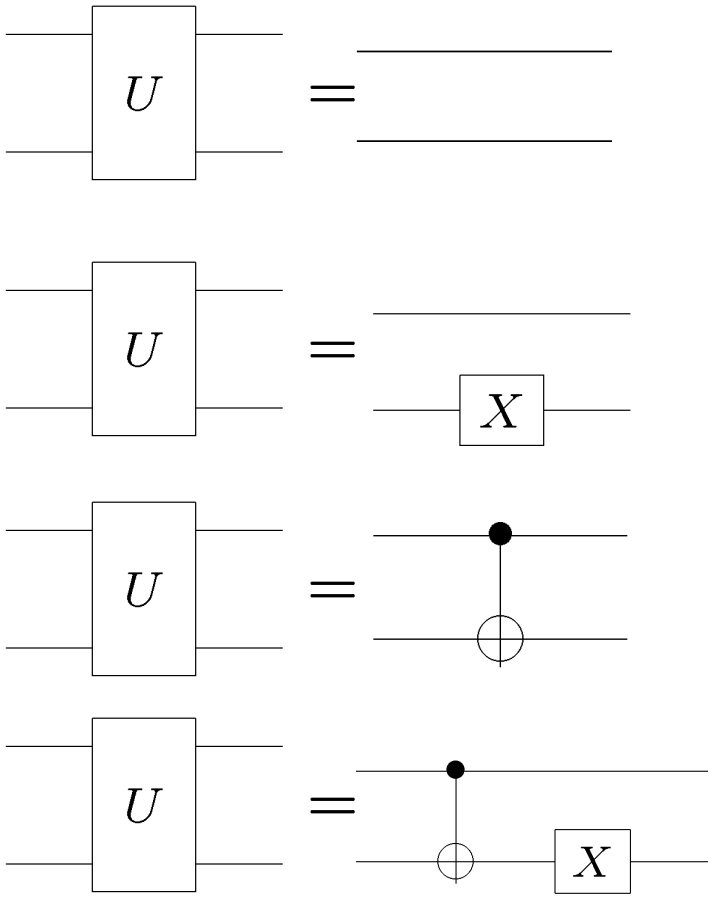


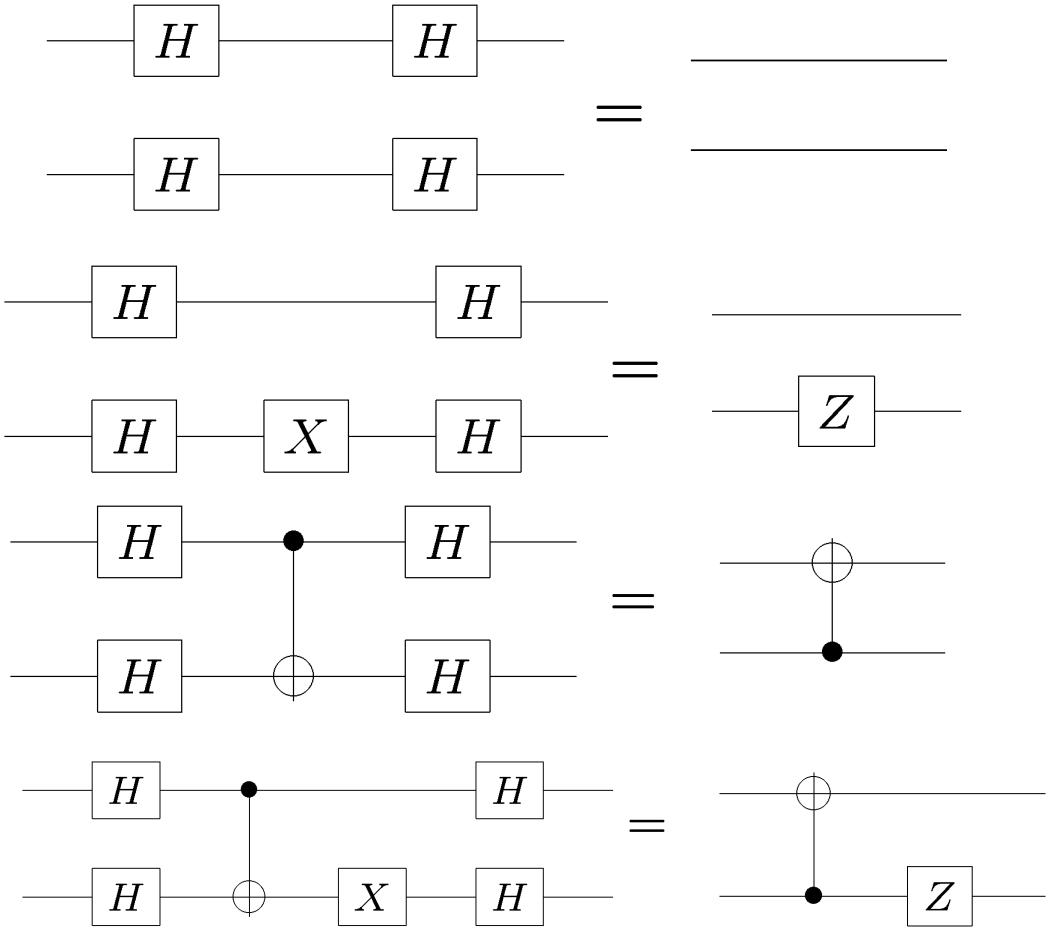
Deutsch's (Quantum) Problem:
How many times do we have to use these quantum gates to determine whether we are given the first two or the second two?

Quantum Deutsch

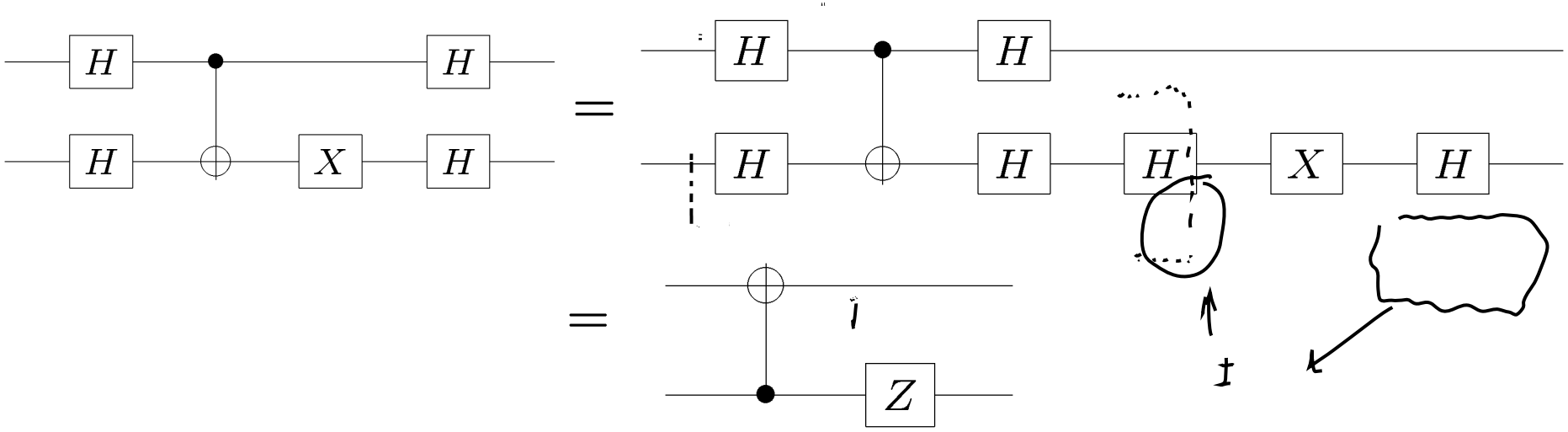
What if we perform Hadamards before and after the quantum gate:



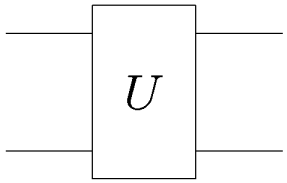


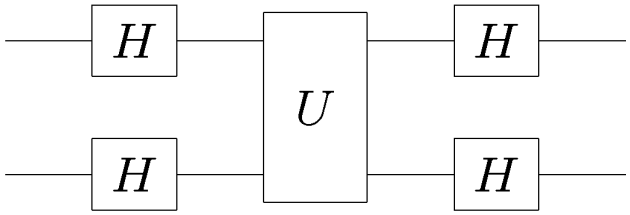


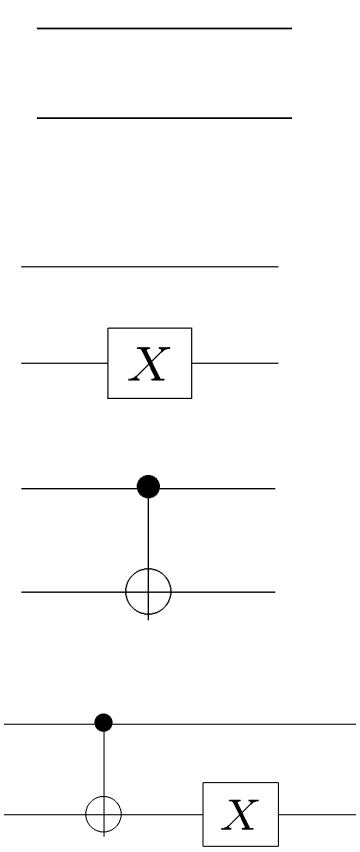
That Last One

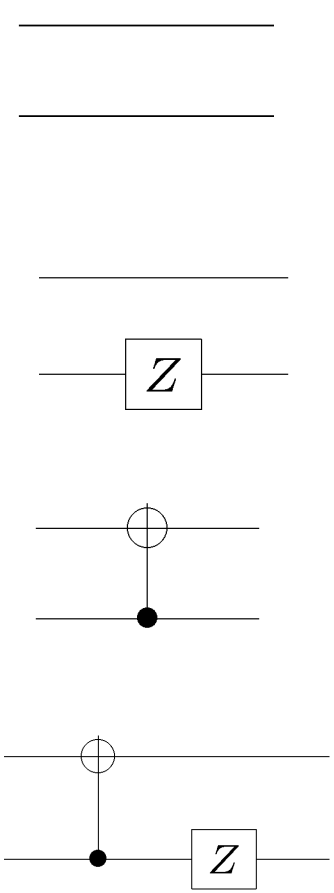


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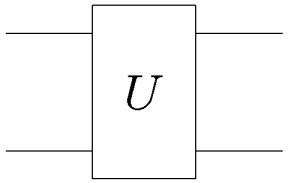


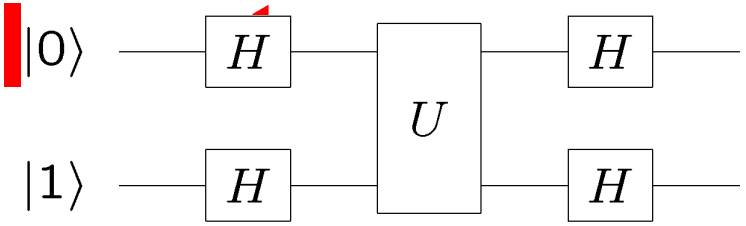


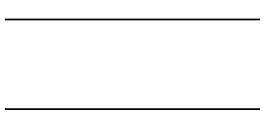


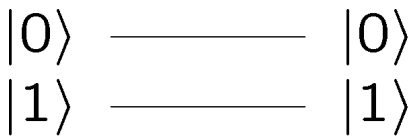


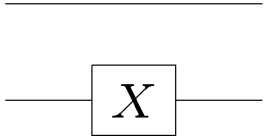
Some

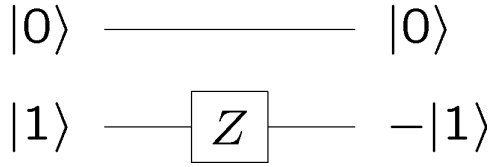


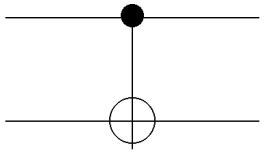


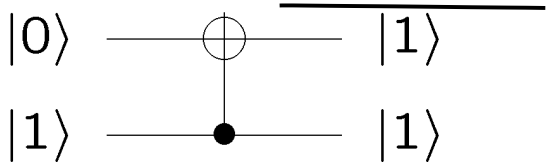


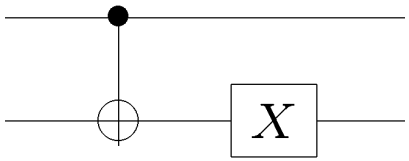


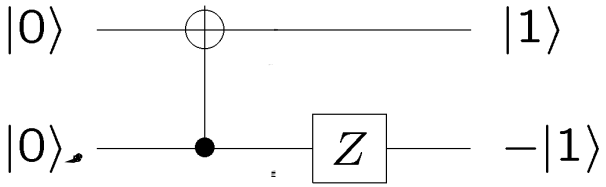




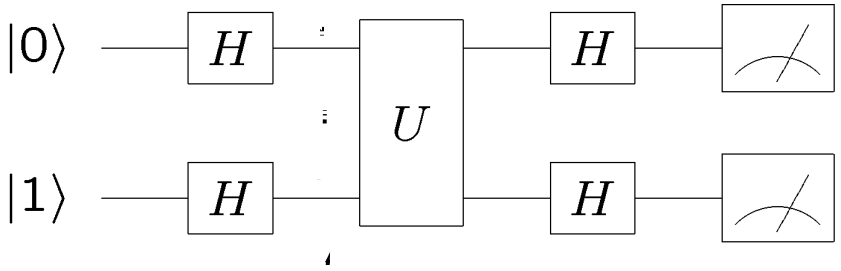


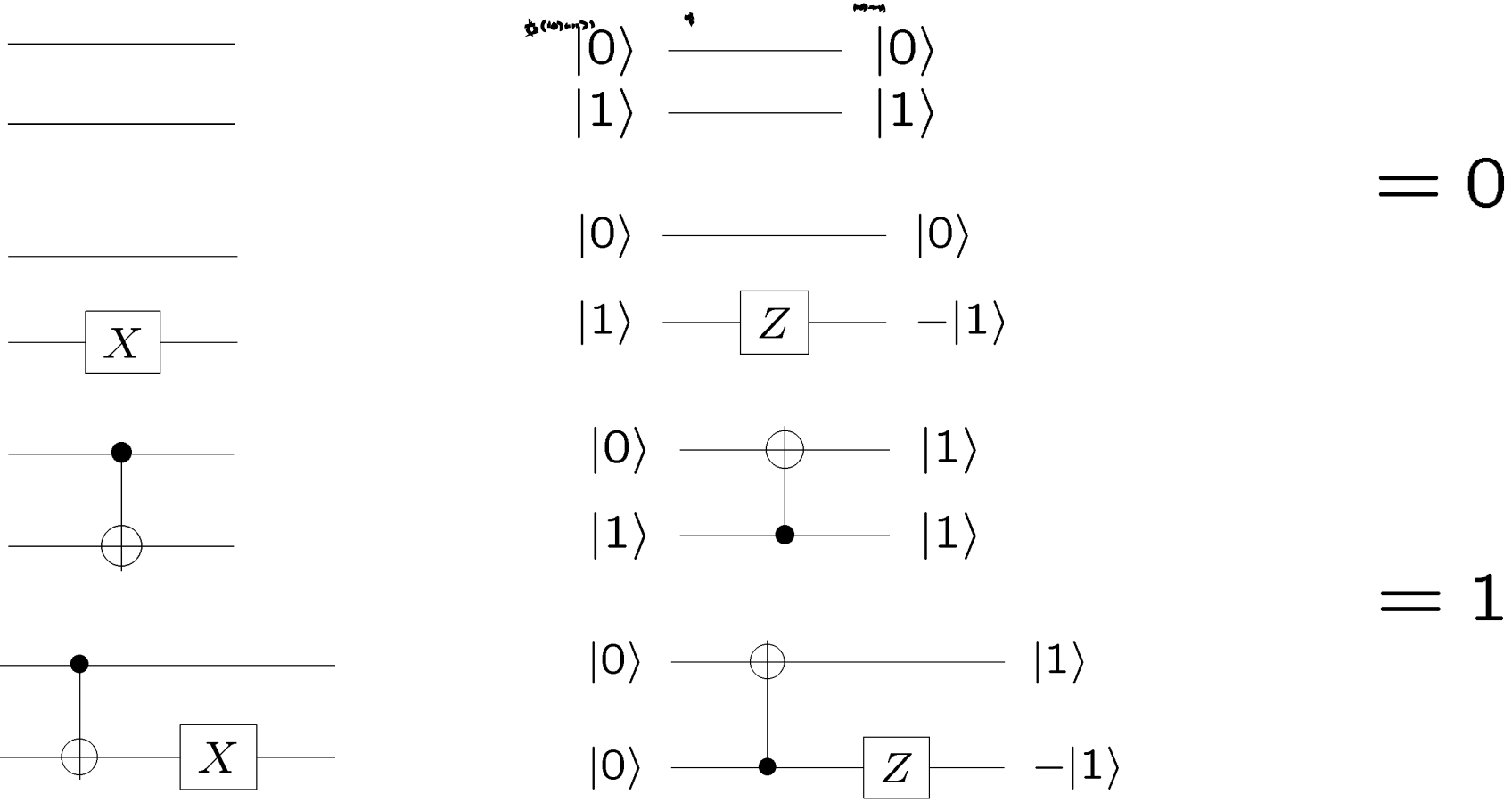




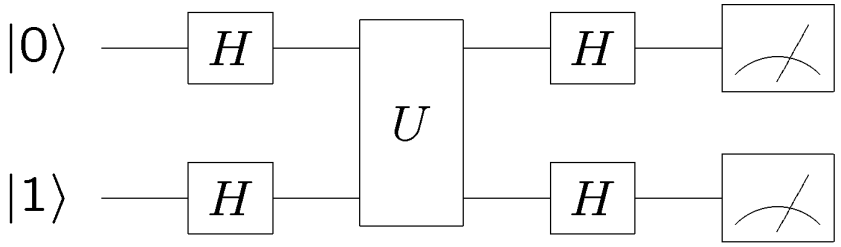


Quantum Deutsch





Quantum Deutsch



By querying with quantum states we are able to distinguish the first two (constant) from the second two (balanced) with
only one use of the quantum gate!

Two uses of the classical gates
Versus
One use of the quantum gate

first quantum speedup (Deutsch, 1985)